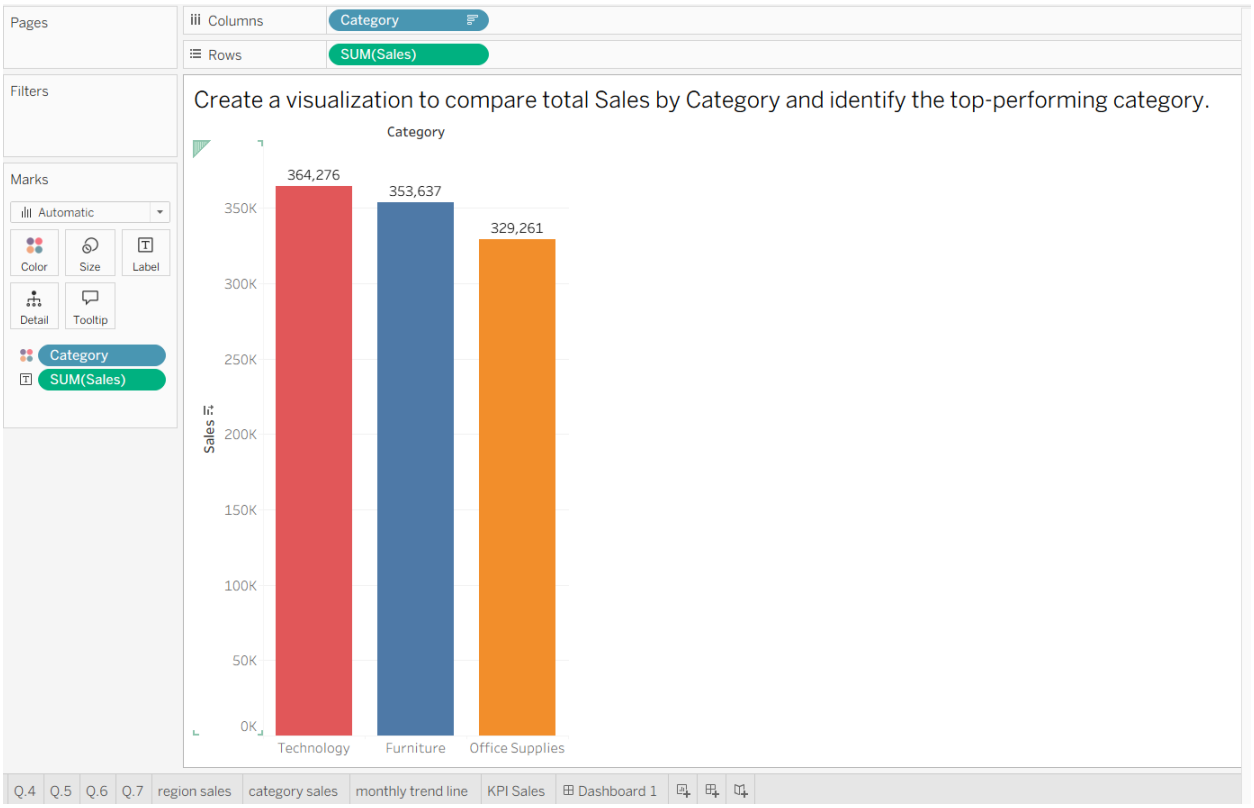
Tableau Visualization Assignment

Task 1.

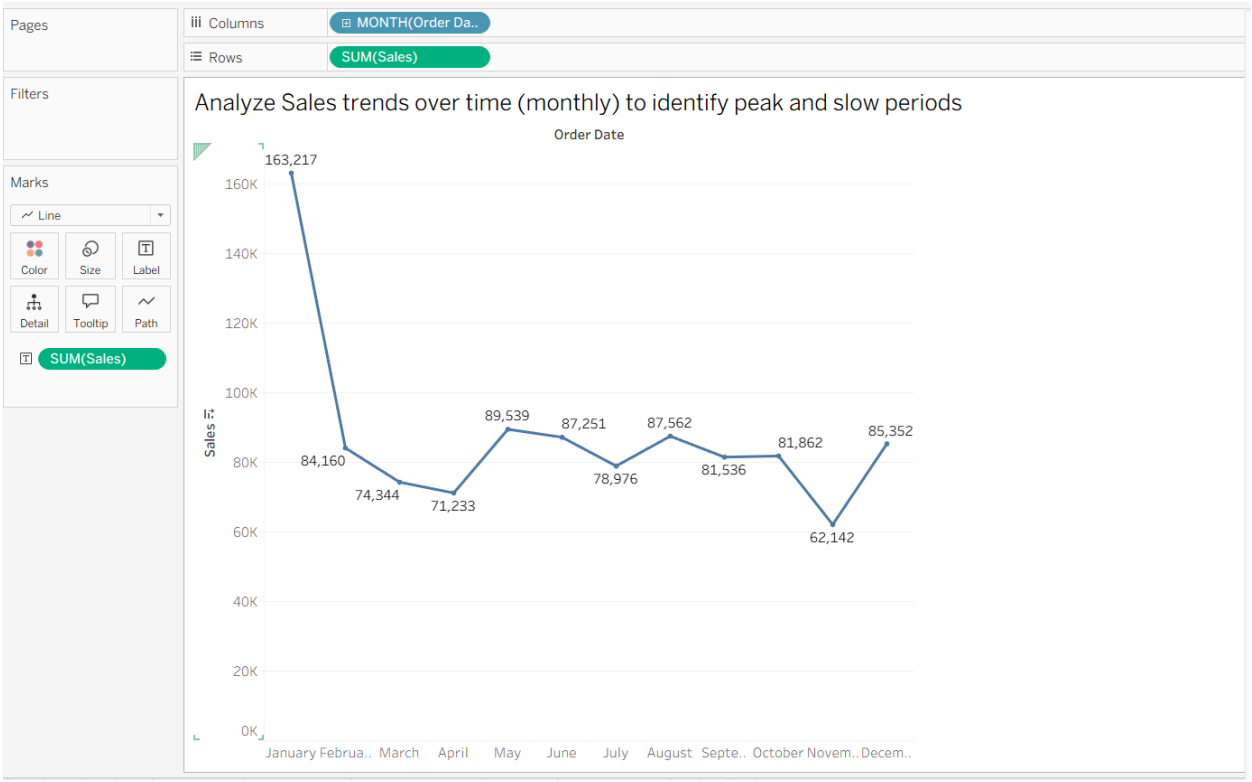
Ans : So, sales analysis shows that technology leads in generating revenue while office supplies needs to be improved in order to generate revenue.



Task 2.

Ans : The monthly sales trend fluctuates over the year but sales peak during January months.

So, according to me, businesses holders should improve sales strategies for the rest months so sales keep consistent and growing.

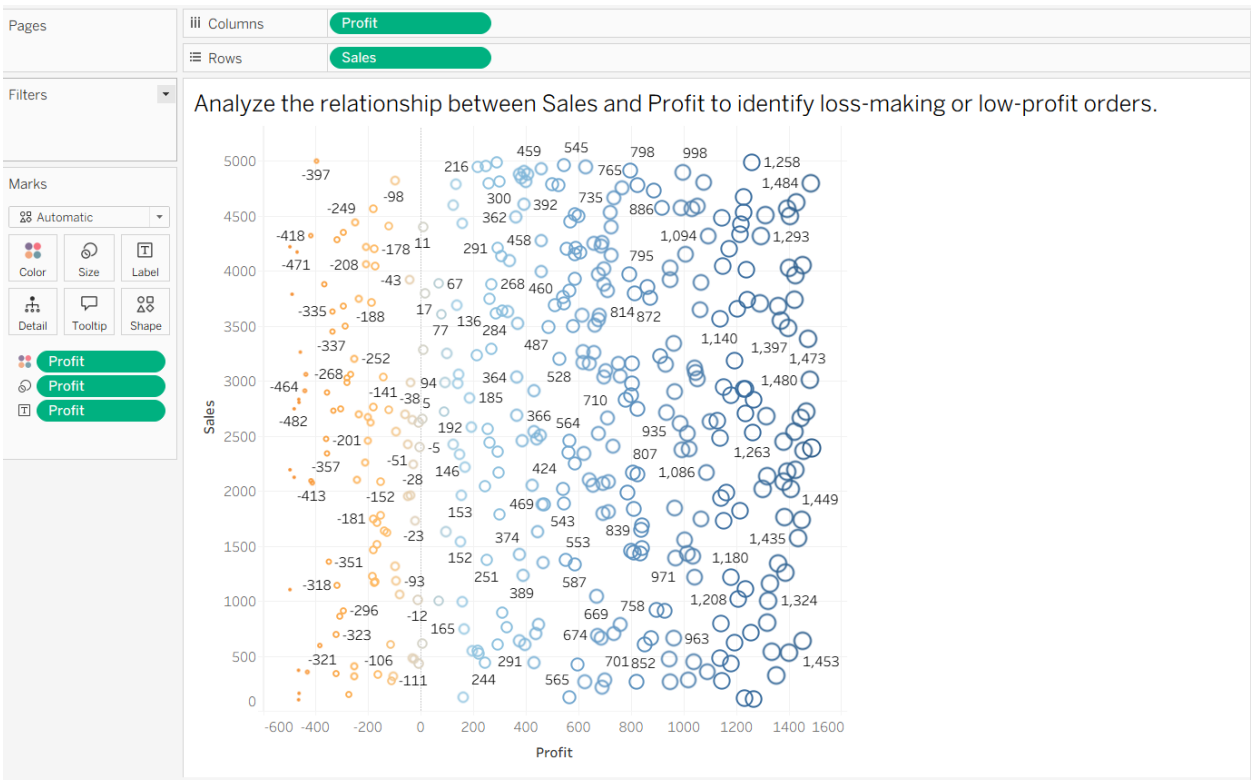


Task 3.

Ans : So, the highest contributing region is west and I think regional manager should improve sales in other remaining regions.

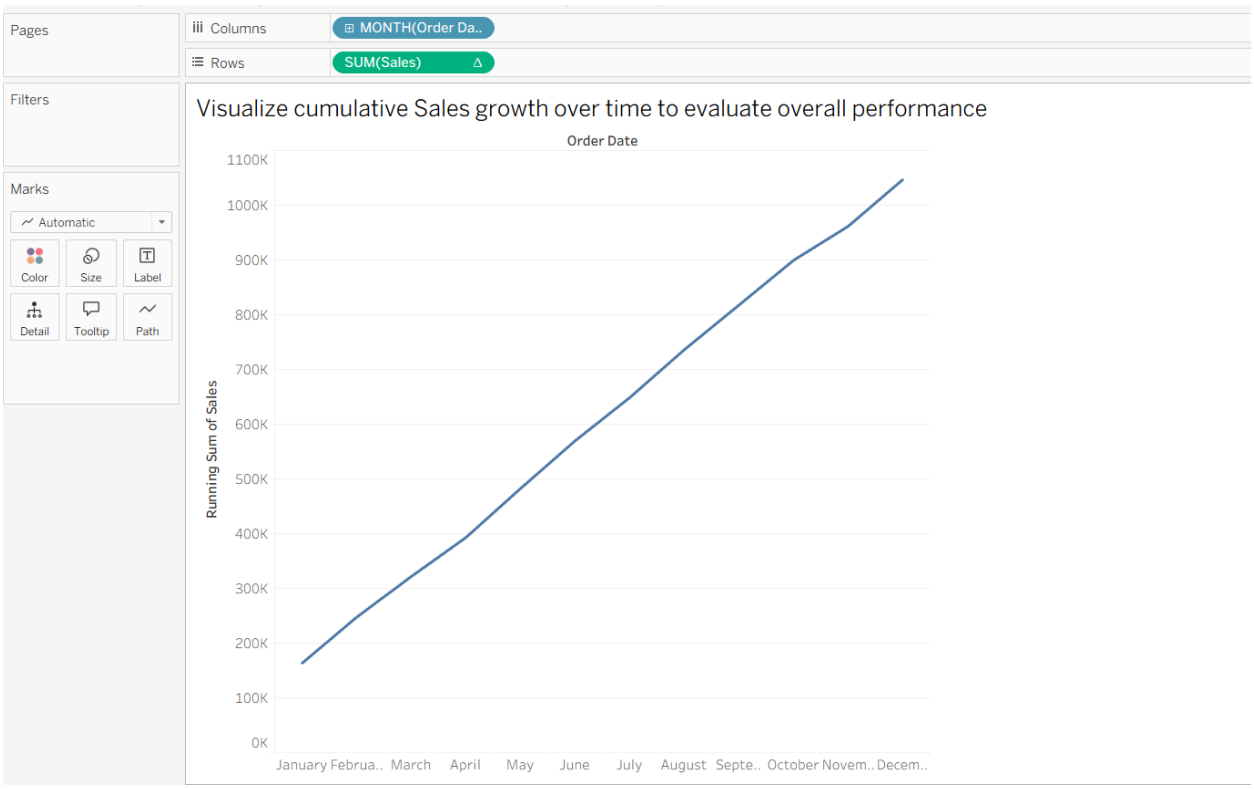
Task 4.

Ans : Here, I can see that the most sales are profitable while some transactions are not profitable despite sales because it often happens because of week discount and pricing strategies. So, it needs to be improved.



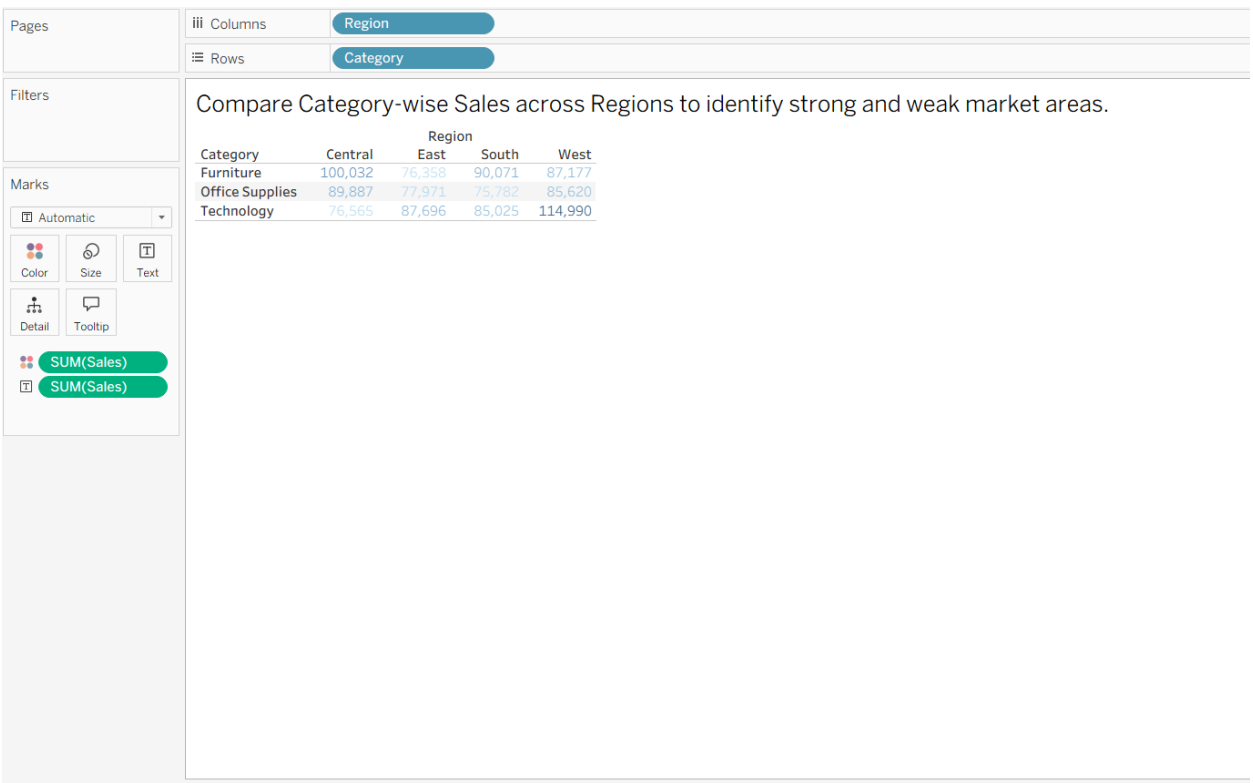
Task 5.

Ans : The cumulative sales trend shows a consistent upward growth over time, indicating overall positive business performance and increasing revenue growth across months.



Task 6.

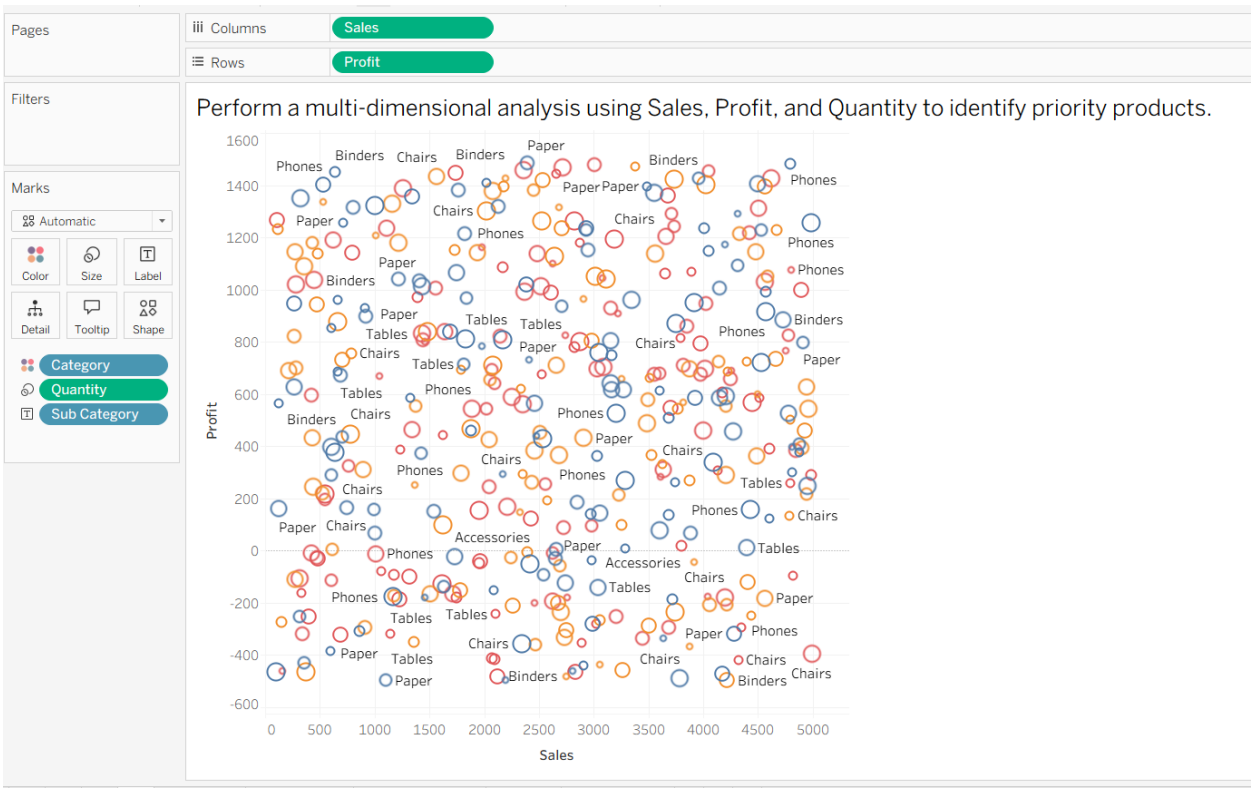
Ans : The analysis shows that the West region is the strongest market, driven significantly by high Technology sales. Central region also performs well, particularly in Furniture, while East region shows comparatively weaker sales across categories. South region demonstrates moderate and balanced performance.



	Region			
Category	Central	East	South	West
Furniture	100,032	76,358	90,071	87,177
Office Supplies	89,887	77,971	75,782	85,620
Technology	76,565	87,696	85,025	114,990

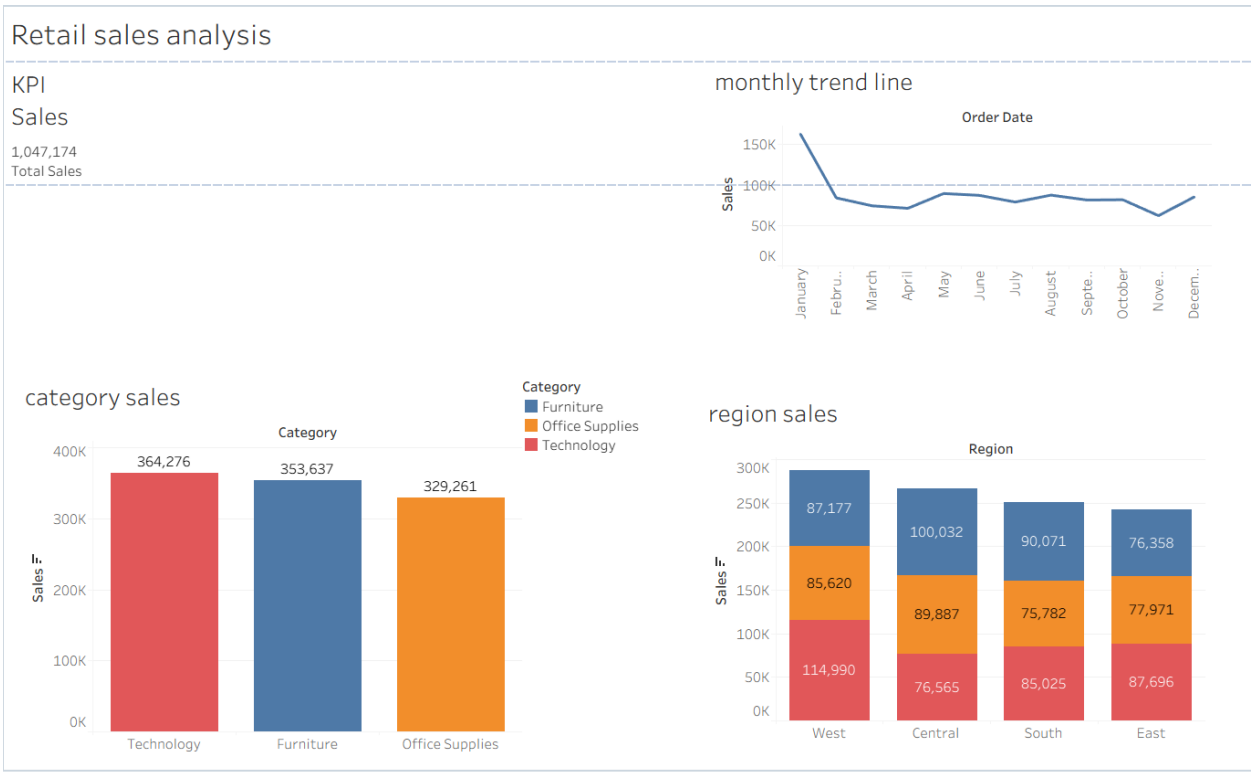
Task 7.

Ans : The multi-dimensional scatter plot shows that Phones emerge as a strong priority product with high sales and strong profitability. Chairs and Tables display mixed performance, with some orders generating losses despite good sales. Paper products show varied results across different transactions.



Task 8: Executive Dashboard Creation Business Need: The CEO wants a single view dashboard that summarizes key insights. Task: Create an interactive dashboard combining at least three visualizations and apply filters such as Region or Category.

Ans :



9. Chart Selection Justification Explain why different charts were used for different business questions in this analysis.

Ans : Different chart types were used in this analysis based on the business question, data type, and the most effective way to present insights clearly. Selecting the right visualization improves understanding and supports better decision-making.

1. Bar Chart for Category-wise Sales Analysis

A bar chart was used to compare total sales across product categories. This chart is ideal for comparing values between categories and quickly identifying the highest and lowest performing categories.

2. Line Chart for Monthly Sales Trends

A line chart was selected to analyze sales trends over time. It effectively shows monthly growth patterns, peak sales periods, and slow-performing months.

3. Bar Chart for Region-wise Sales Contribution

A bar chart was used to compare sales contribution across different regions. It helps rank regional performance and identify the highest contributing region.

4. Scatter Plot for Sales and Profit Relationship

A scatter plot was used to analyze the relationship between Sales and Profit. This chart helps identify profitable orders, low-margin transactions, and loss-making orders.

5. Heat Map for Category-wise Sales across Regions

A heat map was used to compare category performance across multiple regions. The use of color intensity makes it easier to identify strong and weak market areas.

6. KPI Cards for Executive Dashboard

KPI cards were used to display key metrics such as Total Sales, Total Profit, and Total Quantity. These cards provide a quick summary of business performance for management.

Conclusion

Each chart was selected according to the analytical requirement. Bar charts were used for comparisons, line charts for trends, scatter plots for relationships, heat maps for multi-dimensional analysis, and KPI cards for summary metrics. This approach made the dashboard clear, interactive, and useful for strategic decision-making

Q10. Business Insight Summary Summarize two key insights from your dashboard and explain how they can help management make better decisions.

Ans : **Key Insights from Dashboard**

1. Technology Category is the Highest Sales Contributor

The dashboard shows that the **Technology** category generated the highest sales (**364,276**), followed by Furniture and Office Supplies. This indicates strong customer demand for technology-related products.

Management Decision Impact:
Management can increase focus on the Technology category by improving inventory levels, launching targeted promotions, and expanding product variety. This can help maximize revenue and market share.

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2. West Region is the Strongest Performing Market

The region-wise sales analysis shows that the **West** region has the highest total sales contribution compared to Central, South, and East regions. The West region also performs strongly across all product categories.

Management Decision Impact:
Management can prioritize investment in the West region through marketing campaigns, customer retention strategies, and supply chain support. At the same time, lower-performing regions such as East can be reviewed for growth opportunities and improvement plans.

Conclusion

These insights help management identify top-performing categories and regions, allowing them to allocate resources efficiently, improve weaker markets, and make data-driven strategic decisions for future growth.